

Age and growth relationships in species of *Lepomis* in Otsego Lake

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Introduction

Pumpkinseed (*Lepomis gibbosus*), bluegill (*Lepomis macrochirus*), and redbreast sunfish (*Lepomis auritus*) are all members of the genus *Lepomis*, otherwise known as common or true sunfishes. These species are native to Otsego Lake and surrounding water bodies. Otsego Lake, with an area of 6.3 mi² is the largest lake in Otsego County. It is a glacially formed, dimictic lake that is chemically classified as mesotrophic, with some characteristics that would support an oligotrophic classification (Waterfield et al. 2023). Pumpkinseed, bluegill, and redbreast sunfish are among the most plentiful fish in Otsego Lake, and they are both culturally and economically important through recreational angling opportunities they provide. There is little previous investigation into the relationship between age and growth of *Lepomis*, and there is evidence to suggest that populations of some *Lepomis* species in eastern North America are uniquely structured (Copp et al. 2004). There is also evidence to suggest that interspecific interactions may occur in populations of *Lepomis*, and that these interactions can influence population dynamics across species of the genus (Osenberg et al. 1988)

The approximate age of a fish can be estimated by reading its sagittal otolith among other structures. The sagittal otolith of a fish is used to help the fish sense vibrations and plays an important role in their ability to balance and hear (Mirbach et al. 2018). The sagittal otolith is the largest of the 3 pairs of otoliths fishes possess, the other two being the lapillus and asteriscus otoliths. Over a fish's lifetime, the calcium carbonate that comprises the otoliths will continue to build onto the otolith. In the winter, when the fish's metabolism slows down, a dark opaque layer will form on the outside of the otolith. In the warmer months, growth is faster, and the layer that is formed on the otolith is translucent. The opaque bands on an otolith represent years and are known as annuli. Counting the number of annuli can produce an estimate of the approximate age of an individual. Fish can also be aged using structures such as scales or fin rays, however, sagittal otolith aging is the most precise way to age individuals of *Lepomis* (Sakaris et al. 2019, Koch et al. 2019).

The purpose of this study was to use length-at-age data from three species of *Lepomis* in Otsego Lake, NY, to estimate length-at-age of sunfishes within a single statistical model using limited data. To do this, we used otoliths and paired length measurements collected from 2022 through 2024 to estimate mean length at age as well as growth parameters using a multi-species von Bertalanffy growth model within a Bayesian estimation framework. Secondly, we compared growth among species and to regional averages for length at age.

Methods

The individual *Lepomis* used in this study were collected in various ways. Most of the fish used were collected in Indiana style trap nets set by the SUNY Oneonta Biological Field Station (BFS) at Rat Cove and Brookwood Point on Otsego Lake (Hardesty 2024). Anywhere from 2 to 6 fish were collected from trap nets on days when there were fish present in the trap net. These fish were collected on a wide range of days across the summer. Fish taken from the trap nets were brought back to the lab, a measurement of total length (mm) was taken before the fish were dissected, and their sagittal otolith was removed. These fish were also used for parasitological research by BFS researchers (Legname 2024). Some fish were obtained by boat electrofishing during the day in June of 2024. *Lepomis* otoliths collected in previous years by BFS researchers were also used. Sagittal otoliths from *Lepomis* specimens from 2022 and 2023, obtained in similar ways to the otoliths from 2024 were included in this study. These fish were aged along with the individuals from 2024 under the assumption that there had been no drastic changes in the populations.

Whole sagittal otoliths were aged by readers using an Olympus SZ61 stereoscope with an Infinity2 digital camera. Only the light source coming from the bottom of the stereoscope was used so that the annuli on the otoliths would be visible to readers. Ten individuals from each species were aged by both a primary and secondary reader. These 10 individuals were randomly selected from the rest of the sample using R (R Core Team 2023). The ages from both readers were plotted against each other to evaluate possible bias or large discrepancies in perceived ages. In instances where the difference in perceived ages of the primary and secondary reader was greater than one, both readers came together to agree on one age for each fish. The rest of the specimens were aged by only the primary reader.

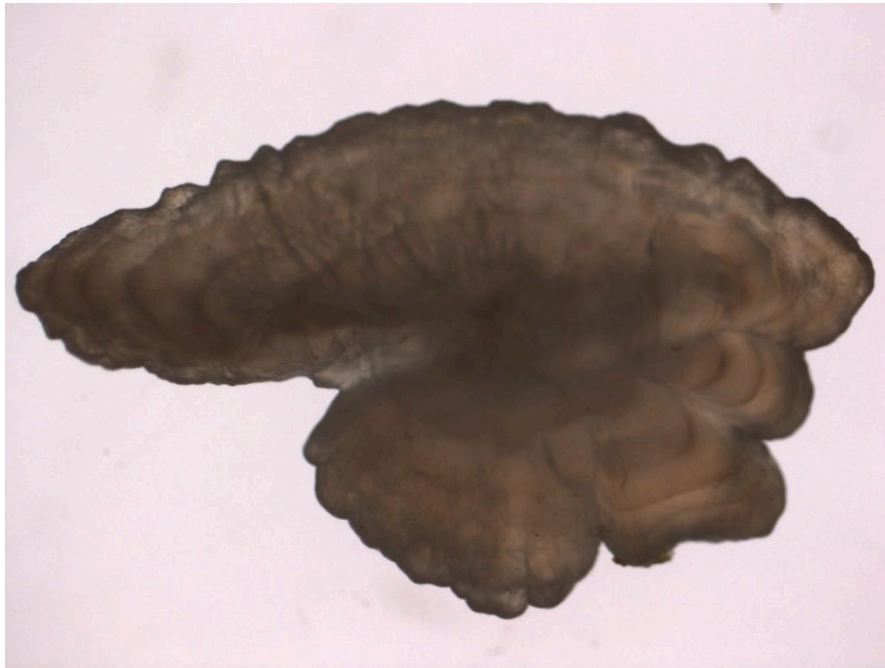


Figure 1: Screen capture of a whole sagittal otolith from a 6-year-old *Lepomis gibbosus* taken using an Olympus SZ61 stereoscope with an Infinity2 digital camera attachment.



Figure 2: Screen capture of a whole sagittal otolith from a 6-year-old *Lepomis macrochirus* taken using an Olympus SZ61 stereoscope with an Infinity2 digital camera attachment.

Statistical Analysis

A multi-species *Lepomis* growth model was made using the von Bertalanffy growth function (von Bertalanffy 1938). The function is used to fit growth curves to species under the assumption that growth slows as fish grow older towards an asymptotic length:

$$L_t = L_{\infty} \times (1 - e^{-k(t-t_0)})$$

Where L_t is individual length at age t . The length a fish would grow to if allowed to grow indefinitely, called the asymptotic length, is represented by L_{∞} . K represents the Brody growth

coefficient, which is the rate at which a fish grows as it approaches asymptotic length. t_0 is the estimated age of an individual when length is zero.

The *brms* package (Bürkner 2017) for R (R Core Team 2023) was used to fit growth models within a Bayesian hierarchical framework. Growth curve parameters were estimated for each species of *Lepomis*, with priors based on global growth curves fitted to all three species of *Lepomis* in this study.

Results

Comparison of ages assigned by two different readers for a subsample of 30 randomly selected fish indicated only slight bias in age assignment among readers. Differences in age assignment by readers were plotted for each individual included in the random sample of 30 *Lepomis* (Figure 3). The slope of the line of best fit for this comparison was -0.04, with the y-intercept being 0.26.

Age and growth data taken from the *Lepomis* in this study were used to create a multi-species von Bertalanffy growth curve and associated parameter estimates that also allowed for species-specific interpretation. Table 1 shows the model outputs for the von Bertalanffy growth function for species of *Lepomis* in Otsego Lake. Asymptotic length for each species, L_∞ , was highest in bluegill, though similar in pumpkinseed, and lowest in redbreast sunfish. The Brody growth coefficient, K , which represents the rate at which a fish approaches asymptotic length, was highest in redbreast sunfish and lowest in bluegill. Despite this variability, the regression coefficients for the multi-species growth model indicated that species did not differ significantly in these estimates due to wide uncertainty intervals.

The von Bertalanffy growth curves are shown for each species of *Lepomis* in Otsego Lake, with age on the x axis and total length on the y axis (Figure 4). As is expected with a fish growth model, growth was greatest early in life, and growth slowed as individuals approached asymptotic length. Model-predicted length-at-age was compared to the average length at age from New York State Department of Environmental Conservation (NYSDEC) Region 4 for each species. These comparisons indicated that bluegill in Otsego Lake grew similarly to the regional average, whereas pumpkinseed grew slightly faster than average. Redbreast sunfish appeared to grow faster than the regional average early in life but reached smaller maximum sizes in Otsego Lake when compared to regional averages.

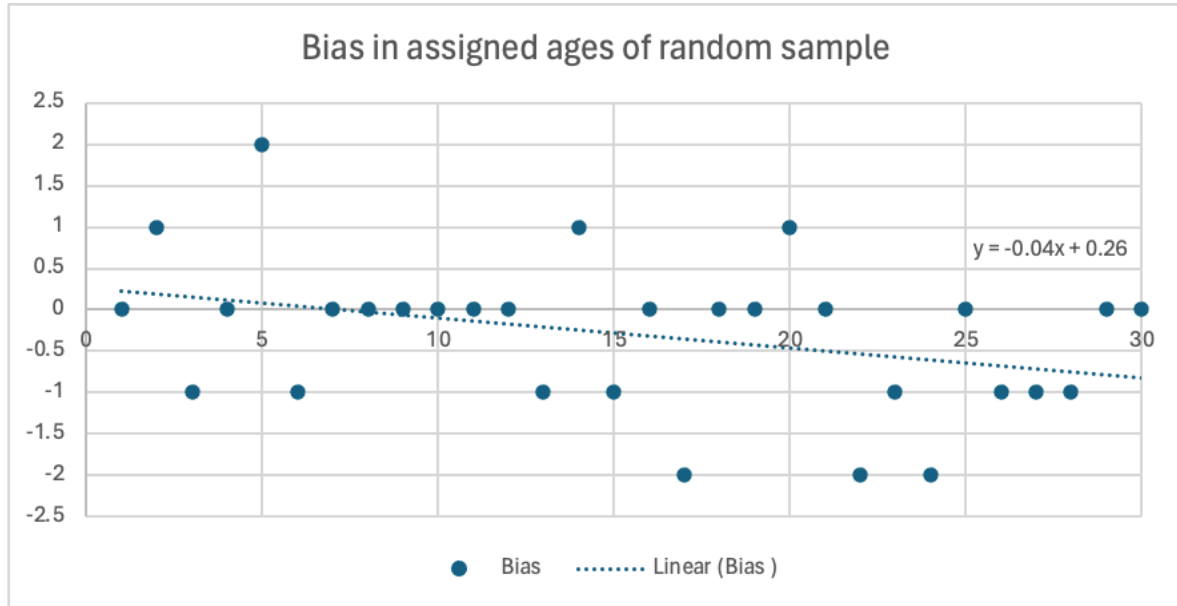


Figure 3: Bias in assigned ages of random sample of *Lepomis*.

species	L_{∞}	K	t_0
Bluegill	239.85	0.295	-0.91
Pumpkinseed	235.10	0.383	-0.45
Redbreast	190.57	0.507	-1.62

Table 1: Model parameters for the von Bertalanffy growth function. L_{∞} is the mean maximum length that a fish would reach if allowed to grow infinitely. K represents the rate of approach to maximum size, and t_0 represents the theoretical age a fish would be at a length of zero.

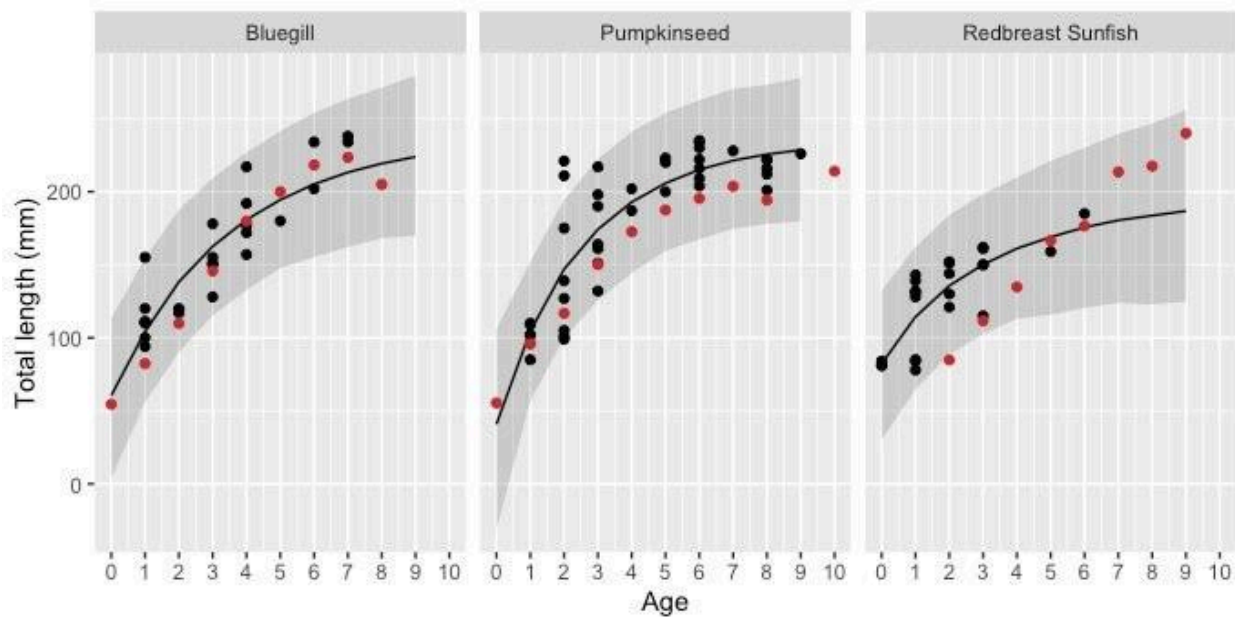


Figure 5: von Bertalanffy growth curves for each species of *Lepomis* in Otsego Lake (black dots) compared to mean length at age from New York State Department of Environmental Conservation Region 4 (red). The black lines are mean length at age predicted from the von Bertalanffy growth function and the gray ribbons are 95% uncertainty intervals.

Discussion

This study was successful in estimating length-at-age and parameters of the von Bertalanffy growth function commonly used in fisheries management for three species of sunfishes in Otsego Lake. The approach used allowed for incorporation of multiple species within a single growth curve to inform estimates for individual species despite limited data. However, more data may be needed to achieve high statistical power for formal hypothesis tests about differences among species.

Bluegill and pumpkinseed had similar L_{∞} though bluegill had the highest asymptotic length. Redbreast sunfish had a much lower L_{∞} than both bluegill and pumpkinseed. Redbreast sunfish however, had the highest growth coefficient, while bluegill had the lowest growth coefficient. This indicates that redbreast sunfish grow to their asymptotic length faster than bluegill, but their asymptotic length is much shorter. For both asymptotic length and growth coefficient, pumpkinseed are similar to bluegill in Otsego Lake but have a smaller asymptotic length and a greater Brody growth coefficient.

Whereas bluegill and pumpkinseed followed similar growth trajectories compared to regional averages, redbreast sunfish from Otsego Lake seem to reach a much smaller asymptotic length compared to the NYSDEC Region 4 average. In general, both bluegill and pumpkinseed were predicted to reach larger sizes and older ages than redbreast sunfish. One reason for this could be the morphometrics of redbreast sunfish. As pumpkinseed and bluegill grow longer, they also increase the vertical profile of their body shape much more than redbreast sunfish. Because of the redbreast sunfish's inability to grow as tall as the other members of *Lepomis*, they could be susceptible to predation from larger gape-limited fish than pumpkinseed or bluegill of similar length. Another reason for the lack of large redbreast sunfish could simply be a failure of the collection methods to detect them. Trap netting and boat electrofishing are both methods used to collect fish that live in shallow water, typically less than 5 feet in depth. Redbreast sunfish, particularly large redbreast sunfish, may live deeper in the water column of Otsego Lake, making them harder to collect with the methods used in this study. One more possible reason that there was a lack of large redbreast sunfish detected in Otsego Lake compared to region 4 is that some lakes in region 4 are specifically managed for panfish, like Goodyear Lake, in Otsego County, which would include *Lepomis*, and thus redbreast sunfish.

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